Functional Document

INDEX

1. Introduction

2. Product Goal

3. Demography (Users, Location)

4. Business Processes

5. Features

    5.1 Feature #1

        1. Description

        2. User Story

6. Authorization Matrix

7. Assumptions

1. Introduction

The Cardiac Arrhythmia Clinical Decision Support System (CDSS) project focuses on revolutionizing edge diagnostics by enabling the direct analysis of physical, artifact-heavy ECG printouts using deep learning. This phase is focused on implementing the core inference pipeline and the Explainable AI (XAI) dashboard to provide clinicians with transparent, highly confident automated diagnoses.

2. Product Goal

The primary goal of this product is to provide a scalable, accessible, and transparent diagnostic tool that mitigates the need for extensive expert-labeled datasets. By utilizing a smartphone-based capture system and XAI heatmaps, the system aims to lower data acquisition costs and support global health accessibility targets, specifically UN SDG 3.

3. Demography (Users, Location)

Users

Target Users: Cardiologists, General Practitioners, Clinical Technicians, Medical Researchers, and System Administrators.

User Characteristics: Varying technical proficiency; requires high transparency in AI decision-making for clinical trust.

Location

Target Location: Global usage, with a specific focus on low-resource clinical settings where traditional digital ECG machines are unavailable.

4. Business Processes

The key business processes include:

User Authentication & Auditing: Process for medical staff to securely log in, ensuring all diagnostic actions and data views are recorded for healthcare regulatory compliance.

ECG Image Capture & Inference: Process for technicians to upload smartphone images of physical ECGs and receive automated arrhythmia classifications via the ResNet18 microservice.

Clinical Review & XAI Analysis: Process for doctors to review the predicted class against the gradient-based saliency heatmap to verify the physiological waveforms the AI focused on.

5. Features

This document outlines the core diagnostic feature of the CDSS platform.

5.1 Feature #1: Automated ECG Arrhythmia Inference & XAI Visualization

Description: The system allows authorized medical personnel to upload an image of a physical ECG printout. The backend ML pipeline processes the image through a contrastive learning model to decouple background noise, classifies the anomaly, and returns a diagnostic result alongside an Explainable AI (XAI) heatmap overlay.

User Story: As a Cardiologist or General Practitioner, I want to upload a smartphone picture of a physical ECG printout so that the system can automatically detect arrhythmias and highlight the exact physiological waveforms it used for its prediction, allowing me to trust and verify the AI's diagnosis quickly and accurately.

6. Authorization Matrix

Define the roles and their corresponding access levels within the CDSS:

Role	Access Level
System Administrator	Full access to user management, role assignment, system settings, and audit logs.
Doctor / Cardiologist	Access to upload ECGs, view inference results, analyze XAI heatmaps, and override automated predictions (Second Opinion).
Clinical Technician	Access to capture and upload physical ECG images; limited access to view final diagnostic results.
Researcher	Access to anonymized historical ECG data, inference logs, and model versioning for label-efficiency analysis.

7. Assumptions

The physical ECG printouts captured via smartphones will have sufficient lighting and clarity for the Image Preprocessing module to function effectively.

The development environment and machine learning infrastructure (specifically GPU resources for the ResNet18 model) will remain stable during the sprint.

Clinicians and users will have a baseline understanding of how to interpret gradient-based XAI heatmaps to validate the model's predictions.